

# ✓ Setup Complete - Mobile CPR Detector

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## Yang Sudah Dilakukan

### 1. ✓ Node.js Upgraded

- **Sebelum:** Node v20.18.3
- **Sekarang:** Node v24.12.0 (LTS terbaru)
- **NPM:** v11.6.2

### 2. ✓ Expo SDK Updated

- Menggunakan **Expo SDK 54** (terbaru)
- Compatible dengan Expo Go app versi terbaru di smartphone

### 3. ✓ Dependencies Fixed

Package yang sudah diupdate:

```
{
  "expo": "~54.0.0",
  "expo-camera": "~17.0.10",
  "expo-status-bar": "~3.0.9",
  "react": "19.1.0",
  "react-native": "0.81.5",
  "react-native-svg": "15.12.1",
  "@tensorflow/tfjs": "^4.11.0",
  "@tensorflow-models/pose-detection": "^2.1.0"
}
```

### 4. ✓ Camera API Updated

- Menggunakan **CameraView** dan **useCameraPermissions** (API terbaru)
- Compatible dengan expo-camera v17

### 5. ✓ Assets Added

- ✓ icon.png
- ✓ splash.png
- ✓ adaptive-icon.png
- ✓ favicon.png
- ✓ models/ folder

## Cara Menjalankan

### 1. Install Dependencies (jika belum)

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```
cd /home/amelia-ocha/Documents/project_izin/mobile-cpr-detector
npm install
```

## 2. Start Expo Dev Server

```
npx expo start -c
```

## 3. Scan QR Code

- **Android:** Buka app **Expo Go**, klik "Scan QR code"
- **iOS:** Buka app **Camera**, arahkan ke QR code

## 4. Grant Permission

Saat app terbuka pertama kali:

1. App akan meminta **Camera Permission** ✓
2. Klik **"Allow"** atau **"Grant Permission"**
3. Kamera akan langsung terbuka! 📷



## Expected Behavior

Saat app berhasil jalan:

### 1. Loading Screen

- "Loading AI Model..."
- "Initializing MoveNet Pose Detection"

### 2. Permission Screen (jika belum granted)

- "x Camera permission required"
- Button "Grant Permission"

### 3. Camera Screen ✓

- Live camera feed terlihat
- Top bar: FPS counter
- Status: "⚠️ Waiting for pose..." atau "CPR DETECTED"
- Bottom controls: "👁️ Show Skeleton" button
- Instructions box di kiri atas



## Troubleshooting

Masalah: Kamera Masih Hitam

### Solusi 1: Force Restart

```
# Stop server (Ctrl+C)
# Clear cache
npx expo start -c
```

### Solusi 2: Reinstall di Device

1. Di Expo Go app, long-press pada project
2. Delete/Remove project
3. Scan QR code lagi

### Solusi 3: Check Permission di Device

- **Android:**
  - Settings → Apps → Expo Go → Permissions → Camera ✓
- **iOS:**
  - Settings → Expo Go → Camera ✓

Masalah: "SDK Version Mismatch"

**Solusi:** Update Expo Go app di smartphone

1. Buka Play Store (Android) atau App Store (iOS)
2. Search "Expo Go"
3. Update ke versi terbaru

Masalah: Metro Bundler Error

**Solusi:** Clean install

```
rm -rf node_modules package-lock.json .expo
npm install
npx expo start -c
```

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## III Performance Notes

Current Limitations:

1. **Frame processing** masih simplified (placeholder)
2. **TensorFlow.js di React Native** memiliki keterbatasan performa
3. Untuk **real pose detection**, perlu implementasi native bridge

Recommended Next Steps:

1. Implement actual camera frame capture
2. Optimize TensorFlow.js backend (webgl/wasm)
3. Consider using **TensorFlow Lite** untuk performa lebih baik

4. Add frame rate throttling (currently placeholder)

## Features yang Sudah Ada

- ✓ Camera view dengan live feed
- ✓ Permission handling
- ✓ UI Overlay (CPR point, skeleton toggle)
- ✓ FPS counter
- ✓ Status indicator
- ✓ Instructions overlay
- ✓ Clean, modern UI

## SOON Next Development Tasks

### 1. Implement actual pose detection

- Connect camera frames to TensorFlow.js
- Process frames at ~5-10 FPS
- Display detected keypoints

### 2. Add CPR calculation

- Calculate compression point from detected pose
- Visualize with marker overlay
- Show confidence score

### 3. Performance optimization

- Optimize tensor operations
- Add frame skipping
- Memory management

### 4. Testing

- Test on different devices
- Test different lighting conditions
- Test different body positions

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## Tech Stack

- **Framework:** React Native + Expo SDK 54
- **Camera:** expo-camera v17
- **AI/ML:** TensorFlow.js + MoveNet Pose Detection
- **Graphics:** react-native-svg
- **Runtime:** Node.js v24.12.0

## Learning Resources

- [Expo Camera Docs](#)
- [TensorFlow.js](#)

- [MoveNet Model](#)
- [React Native](#)

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## ☆☆ Success Checklist

Jika semuanya berhasil, Anda akan lihat:

- ☐ QR code muncul di terminal
- ☐ App terbuka di smartphone
- ☐ Loading screen dengan "Loading AI Model..."
- ☐ Permission request muncul
- ☐ Setelah allow: **KAMERA TERBUKA** ✓
- ☐ UI overlay terlihat (FPS, status, buttons)
- ☐ Console log menampilkan:

```
TensorFlow.js ready  
Backend: webgl  
MoveNet detector loaded
```

**Selamat! 🎉 App sudah berjalan dengan baik!**

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*Last updated: 26 December 2025*

*Node version: v24.12.0*

*Expo SDK: 54*